Short Question Assignment

Question 1: Geometric Intuition.

Suppose you have a transformation matrix A. and a known eignvector v. eigen value $\lambda = -2$.

Geometrically, what happens to the vector v when the transformation A is applied to it?

Solution:

The vector is scaled by a factor of 2, and its direction is completly reversed due to the negative sign. However, it remains perfetly on its original line of span.

Question 2 Matrix powers

Suppose v is an eigenvector of matrix A with an eigenvalue of $\lambda = 3$.

You need to apply the transformation A to

the vector v exactly five times in a row.
What is the first way to compute the result vector without multiply matrix?

Solution: v is the eigenvector

$$A^5 v = \lambda^5 v$$

you simply calculate $3^5 v = 243 v$.

Qustion 3: The Determinant Connection.

A fundamental rule of Linear Algebra is that the product of all eigenvalue of a matrix is equal to it determinint. ($\det(A) = \prod \lambda_i$).

If a 3X3 matrix has eigenvalues $\lambda_1 = 4$, $\lambda_2 = 1.5$, $\lambda_3 = 0$, is the matrix invertable? explain.

Solution: No its not invertable. The $\det(A) = 0$.
Geometricly, one eigenvalue is 0, the transformation

the 3D space to 2D space. plane. making it impossibe reverse the transformation.

Question 4: PCA Intuition.

What exactly does a large eigenvalue represent in the content of a dataset's covariance matrix?

Solution: A large eigen value represent large amount of variance capture long eigenvector direction. they hold most descriptive power reganding the dataset.

Qustion 5: Eigendecomposition Limits.

The eigendecomposition of a matrix A is written as $A = Q \Lambda Q^{-1}$

True or False: Every square matrix can be factored using eigendecomposition. If false state the

Speific Condition a matrix must meet in order for the matrix Q^{-1} to exist.

<u>Solution</u>: False: To be decomposed into $Q\Lambda Q^{-1}$, an $n \times n$ matrix must be exactly n linearly independent eigen vectors. If it does not, the eigenvector matrix Q will not have full rank, and therefore Q^{-1} will not exist.

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